Dashboard Module – Technical Documentation

> **Project:** TaskFlow – Intelligent Task Allocation & Tracking System
> **Module:** Dashboard · **Child Modules:** Overview (Administrator / Manager / Employee dashboards), Analytics
> **Version:** 1.0 · **Last Updated:** 31-Oct-2026
> **Backend:** ASP.NET Core Web API (.NET 10) · **Frontend:** Angular · **Database:** SQLite (EF Core)
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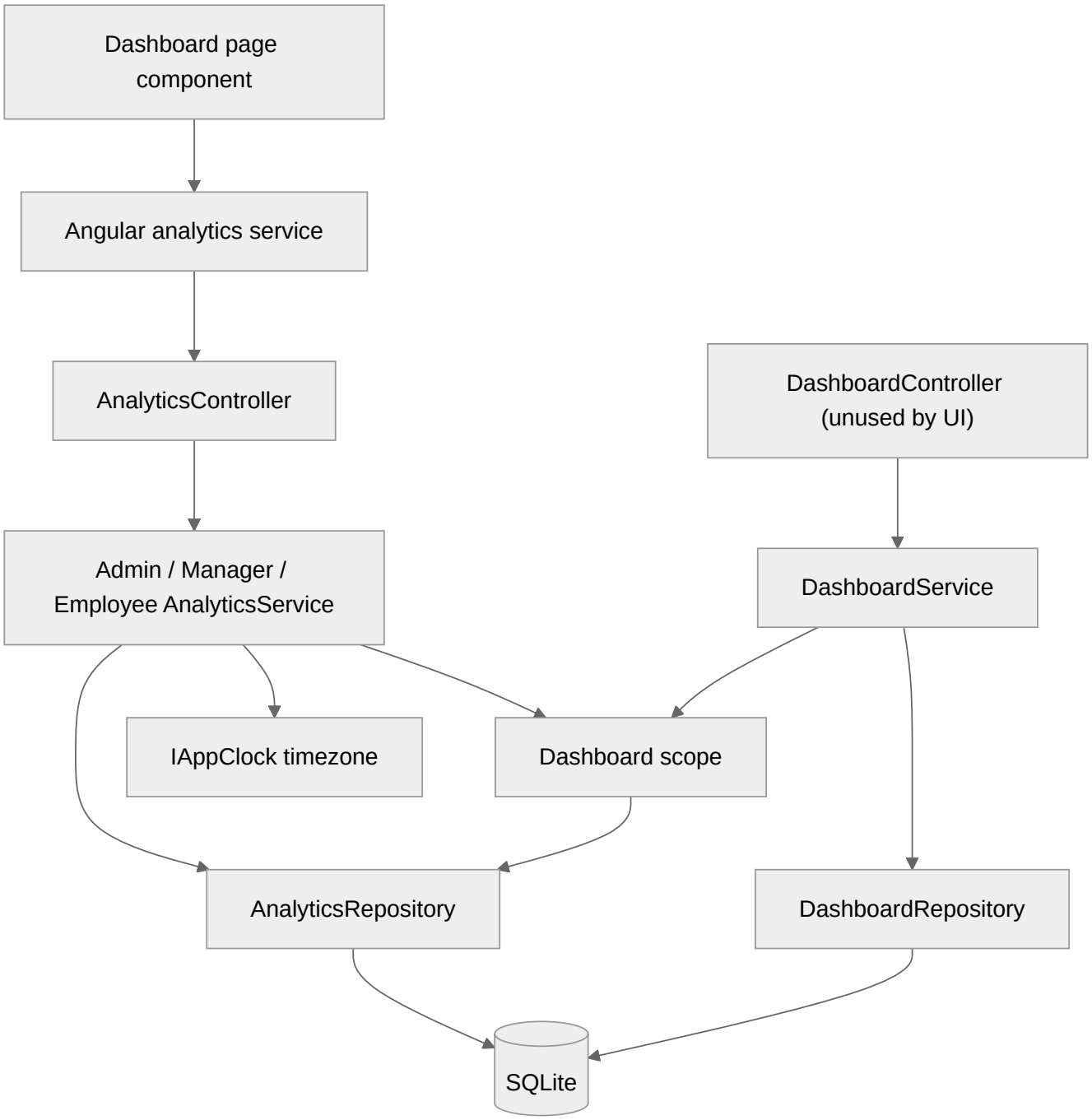
1. Technical Overview

The Dashboard module is read-only. It writes nothing to the database; it aggregates the `Tasks`, `TaskRecipients`, `Users`, `Departments`, `Projects` and link tables into counts for three audiences:

- **Overview** – `DashboardController` + `DashboardService` + `DashboardRepository`: single-query counter endpoints for the Administrator, Manager and Employee (`AdminDashboardDto`, `ManagerDashboardDto`, `EmployeeDashboardDto`). These endpoints exist but no Angular screen calls them today. Details: `02.1-Overview.md`.
- **Analytics** – `AnalyticsController` + `AdminAnalyticsService` / `ManagerAnalyticsService` / `EmployeeAnalyticsService` + `AnalyticsRepository`: the endpoints the three dashboard pages actually use. A **scope** service (`AdminDashboardScope`, `ManagerDashboardScope`, `EmployeeDashboardScope`) turns the caller and filter into an `AnalyticsTaskSet`, and every count runs through one bucket definition (`TaskBucketRules`, `DeadlineBucketRules`) so the dashboards and the Task List drill-down agree. Details: `02.2-Analytics.md`.

Day boundaries ("today", "due today", "next 7 days", date ranges) are computed in the application timezone from `IAppClock.GetTimeZoneAsync` and converted to UTC instants before querying.

2. Architecture Flow



3. Backend File Structure

Module-level summary; the per-child file lists are in the child documents.

Layer	Path	Files / Owner
Controller	`TaskFlow.Api/Controllers/Dashboard/`	`DashboardController.cs` (02.1), `AnalyticsController.cs` (02.2)
Service	`TaskFlow.Api/Services/Dashboard/Overview/`	02.1

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Layer	Path	Files / Owner
Service	`TaskFlow.Api/Services/Dashboard/Analytics/` (`Admin/`, `Manager/`, `Employee/`, `Scopes/`)	02.2 (`IManagerDashboardScope` is also used by 02.1)
Repository	`TaskFlow.Api/Repositories/Dashboard/Overview/`	02.1
Repository	`TaskFlow.Api/Repositories/Dashboard/Analytics/`	02.2
DTOs	`TaskFlow.Api/Dtos/Dashboard/` (`DashboardDto.cs`), `TaskFlow.Api/Dtos/Dashboard/Analytics/`	02.1 / 02.2
Validators	–	No FluentValidation validator; filters are validated in `AnalyticsFilterResolver` and `EmployeeAnalyticsService.ParsePeriod` (02.2)
Shared rules	`TaskFlow.Api/Common/Rules/`	`TaskBucketRules.cs`, `DeadlineBucketRules.cs`, `ManagerTeamThresholds.cs` (and `ReminderCalculator` for local-day maths) – shared
Exceptions	`TaskFlow.Api/Common/Exceptions/`	`AnalyticsExceptions.cs` (02.2)
Entities (read only)	`TaskFlow.Api/Models/Entities/`	`Task.cs`, `TaskRecipient.cs`, `User.cs`, `Department.cs`, `UserDepartment.cs`, `DepartmentManager.cs`, `Project.cs`, `ManagerProject.cs`, `UserProjectAllocation.cs` – shared
Enums	`TaskFlow.Api/Models/Enums/`	`TaskStatus.cs`, `TaskPriority.cs`; `AnalyticsDateBasis` and `EmployeeAnalyticsPeriod` are declared in the DTO files
Authorization	`TaskFlow.Api/Authorization/Policies/`, `TaskFlow.Api/Common/Security/`	`RequirePermissionAttribute.cs`, `PermissionPolicy.cs`, `PermissionKeys.cs`, `PermissionCatalog.cs`, `EffectivePermissions.cs` – shared
DI registration	`TaskFlow.Api/Config/`	`ServiceRegistration.cs`, `RepositoryRegistration.cs` (all scoped) – shared

4. Frontend File Structure

Module-level summary; the per-child file lists are in the child documents.

Type	Path	Purpose / Owner
Page components	`src/app/modules/dashboard/pages/`	`admin-dashboard`, `manager-dashboard`, `employee-dashboard` (02.1; they load data through the 02.2 services)

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Type	Path	Purpose / Owner
Route / Guard	`src/app/modules/dashboard/dashboard.routes.ts`	`admin` – `permissionGuard` with `dashboard.view.organization`; `manager` – `permissionGuard` with `dashboard.view.team`; `me` – no guard (shell `authGuard` + `mfaSetupGuard` only) (02.1)
Landing route / Menu	`src/app/shared/auth/role-routes.ts`, `src/app/layouts/nav-items.ts`	`dashboardRouteFor(role, permissions)`; single "Dashboard" item (02.1)
Analytics components, filter bar, charts	`src/app/modules/dashboard/analytics/components/`, `src/app/modules/dashboard/components/`	02.2
API services	`src/app/modules/dashboard/analytics/services/` (02.2); `src/app/modules/dashboard/services/dashboard.service.ts` (02.1, unused)	–
Models / Utilities	`src/app/modules/dashboard/analytics/models/`, `src/app/modules/dashboard/models/`, `src/app/shared/utils/analytics-filter.ts`, `src/app/shared/utils/drill-down.ts`, `src/app/modules/dashboard/utils/`	02.2 (`dashboard.model.ts` is 02.1)
Permission constants	`src/app/shared/auth/permissions.ts`	`DashboardViewTeam`, `DashboardViewOrganization`, `DashboardAnalytics`, `DashboardFilters` – shared

5. Child Modules

#	Child Module	Main endpoints (count)	Key permissions	Document
02.1	Overview	`GET /api/v1/dashboard/admin`, `/manager`, `/employee` (3)	`dashboard.view.organization`, `dashboard.view.team`, `dashboard.view.own`	`02.1-Overview.md`
02.2	Analytics	`GET /api/v1/dashboard/admin/analytics`, `/admin/analytics/filters`, `/manager/analytics`, `/manager/analytics/filters`, `/employee/analytics` (5)	`dashboard.analytics.view`, `dashboard.filters.use` (with `dashboard.view.organization` / `dashboard.view.team`)	`02.2-Analytics.md`

The child documents are in the `02-Dashboard/` folder next to this file.

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6. Database Design

The module owns no table and runs no migration; it reads the tables below (SQLite, configured in `Persistence/AppDbContext.cs`). Column details are in each child document.

Table	Used by	Purpose in the module
`Tasks`	02.1, 02.2	Every count, list and trend
`TaskRecipients`	02.1, 02.2	A recipient counts as an owner of the task
`Users`	02.1, 02.2	People, names, active flag, managers (`RoleId = 2`)
`Departments`	02.2	Department rows and names
`UserDepartments`	02.1 (team resolution), 02.2	Department membership
`DepartmentManagers`	02.1 (team resolution), 02.2	Departments a manager runs
`Projects`	02.1, 02.2	Project rows and names
`ManagerProjects`	02.2	Projects assigned to a manager
`UserProjectAllocations`	02.1 (team resolution), 02.2	Project membership; project-manager flag widens a manager's team
`RolePermissions` / `Permissions`	02.2	Used to exclude roles holding `tasks.view.all` from the "Employees" count

Relationships

- `Tasks.AssignedToUserId → Users.Id` (FK, Restrict)
- `Tasks.ProjectId → Projects.Id` (FK, Restrict)
- `TaskRecipients.TaskId → Tasks.Id`, `TaskRecipients.UserId → Users.Id` (FK, Cascade)
- `UserDepartments.UserId → Users.Id`, `UserDepartments.DepartmentId → Departments.Id` (FK, Cascade)
- `DepartmentManagers.ManagerUserId → Users.Id`, `DepartmentManagers.DepartmentId → Departments.Id` (FK, Cascade)
- `ManagerProjects.ManagerUserId → Users.Id`, `ManagerProjects.ProjectId → Projects.Id` (FK, Cascade)
- `UserProjectAllocations.UserId → Users.Id`, `UserProjectAllocations.ProjectId → Projects.Id` (FK, Cascade)

7. Security & Authorization

- **Authentication:** JWT bearer token + refresh-token cookie; the caller id is read from `ClaimTypes.NameIdentifier`.
- **Authorization:** one policy per catalogue key named `perm:<key>` (`PermissionPolicy.NameFor`), registered in `Config/CrossCuttingRegistration.cs` as `RequireClaim(AuthClaims.Permission, key)`; permission claims are refreshed from the database on every request. Keys are defined in `PermissionKeys.cs` / `PermissionCatalog.cs`. `EffectivePermissions.Expand` adds lower scope

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levels (`dashboard.view.organization` ⇒ `dashboard.view.team`) and maps legacy keys (`dashboard.admin` ⇒ `dashboard.view.organization`, `dashboard.manager` ⇒ `dashboard.view.team`).

- **Data scope:** Organization – `AdminDashboardScope`; Team – `ManagerDashboardScope` (team + project scope checks, 403 outside scope); Own – `EmployeeDashboardScope` (session user only, no subject parameter accepted).
- **Frontend:** `permissionGuard` on `/admin` and `/manager` redirects to `dashboardRouteFor(...)` when the key is missing; `canFilter()` hides the filter bar without `dashboard.filters.use`. There is no permission directive on the dashboard cards.
- **Denied actions:** `DeniedActionAuditHandler` logs a warning (user id, method, path, policy) for every 403 from a policy.

Permission Key	Endpoint(s)	UI Element Controlled
`dashboard.view.own`	– (Always On, not stored)	`/me` route has no guard; default landing page
`dashboard.view.team`	`GET /api/v1/dashboard/manager`, `GET .../manager/analytics`, `GET .../manager/analytics/filters`	`/manager` route guard; landing for custom roles
`dashboard.view.organization`	`GET /api/v1/dashboard/admin`, `GET .../admin/analytics`, `GET .../admin/analytics/filters`	`/admin` route guard; landing for custom roles; "Administrator" note on Manager Dashboard
`dashboard.analytics.view`	`GET .../admin/analytics`, `GET .../manager/analytics`, `GET .../employee/analytics`	None in the UI (missing key shows the load-error banner)
`dashboard.filters.use`	`GET .../admin/analytics/filters`, `GET .../manager/analytics/filters`	Filter bar on Administrator and Manager dashboards

8. Validations & Error Handling

Shared rules only; filter and period validations are in `02.2-Analytics.md`.

Rule	Layer	HTTP Code	Error Message
Caller id claim missing or not numeric	Controller	401	–
No permission	Policy	403	Forbidden
Unhandled error	`GlobalExceptionHandler`	500	Something went wrong. Please try again.

Errors are returned by `Middleware/GlobalExceptionHandler.cs` as `application/problem+json` (`ProblemDetails`, message in `detail`).

9. Integrations / Side Effects

- **Notifications:** None – the module only reads.
- **Email:** None.

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- **Activity Log:** None written. Refused requests are logged by `DeniedActionAuditHandler`.
- **Task List drill-down:** from the Analytics charts – see `02.2-Analytics.md`.
- **Timezone:** `IAppClock` (Settings module) – stored application setting, falling back to `Application:Timezone`; unknown ids fall back to UTC.
- **AI Assistant:** No Assistant tool or knowledge file references the dashboard.

10. Configuration

Key (`appsettings.json`)	Purpose	Default
`Application:Timezone`	Fallback application timezone for "today" and date ranges (overridden by the stored application setting)	`Asia/Kolkata`
`ConnectionStrings:Default`	SQLite database the queries run against	`<secret>`
`Jwt:AccessTokenMinutes` / `Jwt:RefreshTokenDays`	Session lifetime for dashboard calls	`30` / `7`

Code constants (not configurable): default range 30 days, maximum range 3660 days, daily trend up to 31 days, top 10 employees, attention list 5, upcoming list 10, deadline window 7 days, high-workload factor 1.5 and minimum 5 open tasks, department breakdown minimum 3 active members. Each child document lists the constants it uses.

11. Testing

Paths are relative to `backend/tests`. Shared tests only; child-specific tests are in the child documents.

Test Type	File / Location	Covers
Unit	`TaskFlow.UnitTests/Common/TaskBucketRulesTests.cs`	Bucket classification and completion percent (shared rules)

12. Known Limitations / Future Enhancements

- Custom roles cannot have per-card or per-chart permissions; only the five dashboard keys exist.
- .NET target version: confirmed as `net10.0` in `TaskFlow.Api.csproj`.

Gaps found (module-wide)

- **Counting inconsistency:** Overview counters count Cancelled tasks in `Total` (only `IsDeleted` is excluded), while Analytics excludes Cancelled tasks. Overview "Pending" means all open tasks (including in-progress and overdue), whereas Analytics "Pending" excludes them.
- **Team definition mismatch:** the Manager Dashboard team (`GetTeamMemberIdsAsync`: managed departments + projects where the manager is flagged project manager, active users only) differs from the Admin dashboard's Manager filter and "Manager-wise teams" (`DashboardTaskFilters.TeamUserIds` / `DepartmentManagerTeams`: department membership only, inactive users included). Comments and UI text mention "direct reports", but no direct-report relation exists in the code.

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- **Permission matrix vs code:** the matrix lists `dashboard.view.team` as Granted for Administrator; in code the Administrator gets it by scope expansion from `dashboard.view.organization`. The Administrator is routed to `/admin` by role name (`ROLE_ROUTES` in `role-routes.ts`) before permissions are considered.
- **Permission audit files are out of date:** `Artifacts/permission-audit/*.csv` still point to old paths (`features/dashboard/...`, `core/services/dashboard.service.ts`, `backend/src/TaskFlow.Domain/...`); the current code lives in `modules/dashboard/...` and `TaskFlow.Api/...`.

Child-specific gaps are listed in each child document.

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